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ABSTRACT

This article introduces a new semiparametric panel data model that accounts for time-varying coefficients and aligns with recent advancements in factor models featuring nonparametric loading functions. We propose a profile marginal integration (PMI) method to jointly estimate the unknown quantities in a series of easily implementable steps. The asymptotic properties of these estimators are established. Additionally, we provide a hypothesis test to assess the validity of parametric model specifications in applied settings. Simulation studies and an empirical application on mutual fund returns are conducted to evaluate the finite sample performance of the proposed method. The empirical results suggest that traditional parametric methods, which ignore time variation, may lead to invalid inference.

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1. Introduction

Panel data analysis has been widely adopted across disciplines such as economics, finance, biology, and ecology. Although most studies employ parametric models with time-invariant coefficients (Hsiao 2014), this assumption may not hold in practice (e.g., Silvapulle et al. 2017; Cai, Cheng, and Yan 2018), especially for large panels over long time spans. This limitation has motivated a substantial body of literature to consider panel data models with time-varying coefficients:

$$y_{it} = \mathbf{x}_{it}^T \boldsymbol{\beta}_0(\tau_t) + u_{it}, \quad (1.1)$$

for $i = 1, \dots, N$ and $t = 1, \dots, T$, where the elements of $\boldsymbol{\beta}_0(\tau_t)$ are nonparametric functions of the rescaled time $\tau_t = t/T$. Early studies adopt the classic fixed-effects structure for u_{it} and uses either the local constant (e.g., Robinson 2012) or local linear method (e.g., Li, Chen, and Gao 2011; Chen, Gao, and Li 2012) for the estimation of time-varying coefficients.

Recently, Casas et al. (2021) extend this framework by employing the common factor structure for u_{it} . Compared with classical fixed effects, the common factor structure allows unobserved common factors and individual-specific factor loadings to enter the model in a multiplicative form (see, Pesaran 2006; Bai 2009), providing a more flexible way to capture cross-sectional dependence in panel data. For estimation, Pesaran (2006) constructs common correlated effects (CCE) estimators based on factor proxies derived from observable dependent and independent variables. By contrast, Bai (2009) develops an iteration algorithm that alternates between the least squares estimation and the principal component analysis (PCA).

Another strand of literature (e.g., Connor, Hagmann, and Linton 2012; Fan, Liao, and Wang 2016) models factor load-

ings as nonparametric functions of some observable covariates. Specifically, Connor, Hagmann, and Linton (2012) introduce the semiparametric additive factor structure:

$$\boldsymbol{\lambda}(\mathbf{v}_i)^T \mathbf{F}_t = \sum_{j=1}^J \lambda_j(v_{ji}) F_{jt}, \quad (1.2)$$

where $\mathbf{v}_i = (v_{1i}, \dots, v_{Ji})^T$ represents a vector of observable loading covariates. Here, $\boldsymbol{\lambda}(\mathbf{v}_i) = (\lambda_1(v_{1i}), \dots, \lambda_J(v_{Ji}))^T$ and $\mathbf{F}_t = (F_{1t}, \dots, F_{Jt})^T \in \mathbb{R}^{J \times 1}$ contain unobservable loadings and factors, respectively. In relation to the asset pricing literature, this structure connects factor returns (F_{jt}) to security characteristics (v_{ji}) through the nonparametric characteristic-beta function ($\lambda_j(\cdot)$). Unlike nonadditive functions, this additive specification helps in identifying the distinct risk factors that affect asset prices. These include factors such as size, value, momentum, and own-volatility, as revealed by Connor, Hagmann, and Linton (2012). Another application of (1.2) is in modeling economic growth, where factors such as capital, labor, and technological progress contribute to growth. An additive factor model can separate the contributions of these factors, helping to understand their individual effects. This structure is also applicable to other fields such as health economics and climate change economics, as explored by Casas et al. (2021) and Keane and Neal (2020), among others.

For estimation, Connor, Hagmann, and Linton (2012) provide a recursive procedure alternating between the nonparametric kernel estimation and ordinary least squares (OLS), achieving optimal convergence rates. Moreover, Ma, Linton, and Gao (2021) study quantile regression for the model proposed by Connor, Hagmann, and Linton (2012), focusing on the distributional

behavior of daily stock returns. Chen et al. (2023) extend this work by considering quantile estimation for a characteristic-based factor structure, allowing for a different number of factors than the number of loading covariates. Building on Connor, Hagmann, and Linton (2012), Fan, Liao, and Wang (2016) propose a projected PCA estimator that accommodates unobservable heterogeneous loadings. Most existing methods, however, assume time-invariant loading covariates (see, Connor, Hagmann, and Linton 2012; Ma, Linton, and Gao 2021; Chen et al. 2023, among others), which can be restrictive in practice. Notable exceptions include Chen, Roussanov, and Wang (2021) and Park et al. (2009), who study semiparametric factor models without assuming an additive form.

To advance the literature on time-varying panel regression and semiparametric factor models, we consider a semiparametric model with both time-varying coefficients and an additive factor structure:

$$y_{it} = \mathbf{x}_{it}^\top \boldsymbol{\beta}_0(\tau_t) + \sum_{j=1}^J \lambda_j(v_{jit}) F_{jt} + \varepsilon_{it}, \quad (1.3)$$

where $\lambda_1(v_{1it}), \dots, \lambda_J(v_{Jit})$, and $\boldsymbol{\beta}_0(\tau_t)$ are nonparametric functions of observable loading covariates $v_{1it}, \dots, v_{Jit}$, and time τ_t , respectively. Here, $\mathbf{x}_{it}$ is a $p \times 1$ vector of regressors including an intercept term, and ε_{it} is the idiosyncratic error term.

It should be pointed out that $\mathbf{x}_{it}$ and $\mathbf{v}_{it} = (v_{1it}, \dots, v_{Jit})^\top$ capture different characteristics. $\mathbf{x}_{it}$ represents the main regressors or predictors available to the econometrician, and $\mathbf{v}_{it}$ contains the observed covariates that facilitate the specification of the loading functions. Therefore, the choices of $\mathbf{x}_{it}$ and $\mathbf{v}_{it}$ are driven by research questions, and are known in advance. A closely related study is Dong, Gao, and Peng (2021), which considers the case where each $\mathbf{x}_{it}$ is nonstationary and $\mathbf{v}_{it}$ reduces to $\mathbf{v}_i$ in the same way as in (1.2). However, the series based methodology by Dong, Gao, and Peng (2021) cannot accommodate the involvement of the time-varying loading covariates of the form $\mathbf{v}_{it}$ as in model (1.3). Additionally, compared with the time-varying model with an unobservable common factor structure considered by Casas et al. (2021), model (1.3) offers the advantage of incorporating observable loading covariates into the estimation process, which enhances efficiency.

Notably, due to time-varying loading covariates in model (1.3), the PCA-based estimation method cannot be directly applied because it fails to recover common factors with time-varying loadings through eigenanalysis. Consequently, we turn to the marginal integration (MI) method, which is widely employed by the relevant literature to estimate nonparametric additive models. This method offers the benefits of the advantages of closed-form expressions and ease of implementation (see, e.g., Tjøstheim and Auestad 1994; Newey 1994; Gao 2007) for both cross-sectional and time series data. The MI method has also been developed by Gao, Lu, and Tjøstheim (2006) for spatial data analysis. Given that model (1.3) involves both time-varying regression and additive factor components, we extend the MI method and then propose a profile marginal integration (PMI) estimation approach.

In summary, our main contributions are as follows.

- We introduce an extended model of the form (1.3) to account for time-variations in both regression coefficients and additive factor models.

- The proposed PMI method jointly estimates the unknown functions and parameters, and we establish the corresponding asymptotic properties.
- A hypothesis test is provided to examine parametric model specifications in practice.
- An empirical study is conducted to reveal the nonlinear associations between mutual fund returns and fund characteristic variables.

The rest of the article is organized as follows. Section 2 presents the methodology and the corresponding asymptotic properties. Simulation experiments are conducted in Section 3. Section 4 applies the proposed framework to an empirical study on mutual fund returns. Section 5 concludes. For brevity, implementation instructions, additional discussions on the methodology, and proofs for the theorems are provided in Appendices A–C of the online supplement.

Before proceeding further, we define some notation. For a matrix $\mathbf{A}$, $\mathbf{A}^\top$, and $\|\mathbf{A}\|$ stand for its transpose and Frobenius norm, respectively; $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ are the smallest and largest eigenvalues of $\mathbf{A}$; $\mathbf{I}_a$ and $\mathbf{0}_a$ are $a \times a$ identity and null matrices, and $\mathbf{0}_{a \times b}$ is an $a \times b$ matrix of zeros; “ $\circ$ ” denotes element-wise product of two matrices; “ $\xrightarrow{D}$ ” denotes convergence in distribution.

2. The Methodology and Asymptotic Properties

There are several sources of indeterminacy in (1.3) due to the additive form of the J factor components and the multiplicative relationship between the factor and loading function. These indeterminacies require careful consideration, and thus, some identification restrictions are necessary. Without loss of generality, we impose the same identification conditions as those in Connor, Hagmann, and Linton (2012).

Assumption 1 (Identification).

1. For the loading functions, let $E[\lambda_j(v_{jit})] = 0$ and $E[\lambda_j^2(v_{jit})] = 1$ for each j, i , and t .
2. For the factors, let $\bar{F}_j = \frac{1}{T} \sum_{t=1}^T F_{jt} > 0$ for each j .
3. $\mathbf{x}_{it}$ and $\mathbf{v}_{it}$ are distinct and contain nonoverlapping observable variables.

The first condition in Assumption 1.1 requires that $E[\lambda(\mathbf{v}_{it})^\top \mathbf{F}_t] = \mathbf{0}$, which guarantees the identifiability of the additive factor structure from the regression component. The second condition, together with Assumption 1.2, ensures that once we obtain estimates of the product of factors and loadings, the factors and loadings can be individually recovered. Notably, under Assumption 1, it holds that

$$\begin{aligned} E[\lambda(\mathbf{v}_{it}^{(-j)}(v_j))^\top \mathbf{F}_t] &= E\left[\lambda_j(v_j)F_{jt} + \sum_{k=1, k \neq j}^J \lambda_k(v_{kit})F_{kt}\right] \\ &= \lambda_j(v_j)F_{jt} + \sum_{k=1, k \neq j}^J E[\lambda_k(v_{kit})F_{kt}], \\ &= \lambda_j(v_j)F_{jt}, \end{aligned} \quad (2.1)$$

where $\mathbf{v}_{it}^{(-j)}(v_j)$ is defined by fixing the j th element of $\mathbf{v}_{it}$ to a specified value v_j . This formulation motivates the use of the PMI method below, which uses the sample version of (2.1).

2.1. The PMI Method

We begin by introducing notations essential for constructing the estimators. Throughout, let $K(\cdot)$ denote a nonparametric kernel function, and let $h_0, h_1, \dots, h_J$ represent bandwidths. Define $h_{\min} = \min\{h_1, \dots, h_J\}$, $H_J = \prod_{j=1}^J h_j$ and $H_J^{(-j)} = \prod_{k=1, k \neq j}^J h_k$. Let $\mu_2 = \int v^2 K(v) dv$. Define the following vectors

$$\mathbf{y}_t = (y_{1t}, \dots, y_{Nt})^\top, \mathbf{x}_t = (\mathbf{x}_{1t}, \dots, \mathbf{x}_{Nt})^\top,$$

and let $\mathbf{v}_t$ and $\boldsymbol{\varepsilon}_t$ be defined analogously. Lastly, we introduce several matrices:

$$\begin{aligned} \mathcal{W}_t(\mathbf{v}) &= \text{diag}(\mathcal{K}_{1t}(\mathbf{v}), \dots, \mathcal{K}_{Nt}(\mathbf{v})), \\ \mathbf{W}(\tau) &= \text{diag}\left(K\left(\frac{\tau_1 - \tau}{h_0}\right), \dots, K\left(\frac{\tau_T - \tau}{h_0}\right)\right) \otimes \mathbf{I}_N, \\ \mathcal{M}_t(\mathbf{v}) &= \begin{pmatrix} 1 & \dots & 1 \\ \mathcal{V}_{1t}(\mathbf{v}) & \dots & \mathcal{V}_{Nt}(\mathbf{v}) \end{pmatrix}^\top, \end{aligned}$$

where $\mathcal{V}_{it}(\mathbf{v}) = (\mathbf{v}_{it} - \mathbf{v}) \circ \mathbf{h}_J^{-1}$ is the element-wise product of $(\mathbf{v}_{it} - \mathbf{v})$ and $\mathbf{h}_J^{-1}$, $\mathcal{K}_{it}(\mathbf{v}) = \prod_{j=1}^J K(\frac{\mathcal{V}_{jit} - v_j}{h_j})$, $\mathbf{v} = (v_1, \dots, v_J)^\top$, and $\mathbf{h}_J^{-1} = (h_1^{-1}, \dots, h_J^{-1})^\top$.

We then introduce the PMI method, which is implemented in three steps. First, we apply the marginal integration (MI) technique to obtain a closed-form representation for $\boldsymbol{\lambda}(\mathbf{v}_{it})^\top \mathbf{F}_t$ in terms of time-varying regression coefficients $\boldsymbol{\beta}_0(\tau)$. Next, we substitute this representation into the original model and pursue a local linear estimator for $\boldsymbol{\beta}_0(\tau)$. Finally, the loadings and factors are recovered based on the representation obtained in the first step, using the estimated $\boldsymbol{\beta}_0(\tau)$. This profile approach enables the estimation of all nonparametric and parametric unknown quantities ($\boldsymbol{\beta}_0(\tau)$, $\boldsymbol{\lambda}(\mathbf{v})$, $\mathbf{F}_t$) at their optimal convergence rates with closed-form expressions.

Specifically, the following details the three steps:

Step 1 For any given $\boldsymbol{\beta}(\tau_t)$, we write

$$\gamma_t(\mathbf{v} | \boldsymbol{\beta}) := E[y_{it} - \mathbf{x}_{it}^\top \boldsymbol{\beta}(\tau_t) | \mathbf{v}_{it} = \mathbf{v}].$$

Clearly, $\gamma_t(\mathbf{v} | \boldsymbol{\beta}_0) = \sum_{j=1}^J \lambda_j(v_j) F_{jt} := \sum_{j=1}^J \gamma_{jt}(v_j)$, where $\gamma_{jt}(v_j) = \lambda_j(v_j) F_{jt}$. We define the local linear estimator of $\gamma_t(\mathbf{v} | \boldsymbol{\beta})$ as

$$\hat{\gamma}_t(\mathbf{v} | \boldsymbol{\beta}) = \mathbf{s}_t(\mathbf{v})(\mathbf{y}_t - \mathbf{x}_t \boldsymbol{\beta}(\tau_t)), \quad (2.2)$$

where $\mathbf{s}_t(\mathbf{v}) = [1, \mathbf{0}_{1 \times J}] (\mathcal{M}_t^\top(\mathbf{v}) \mathcal{W}_t(\mathbf{v}) \mathcal{M}_t(\mathbf{v}))^{-1} \mathcal{M}_t^\top(\mathbf{v}) \mathcal{W}_t(\mathbf{v})$. In view of (2.1), for each j we construct the MI following estimator (given $\boldsymbol{\beta}(\tau)$) for $\gamma_{jt}(v_j)$:

$$\hat{\gamma}_{jt}(v_j | \boldsymbol{\beta}) = \frac{1}{N} \sum_{n=1}^N \hat{\gamma}_{nt}(\mathbf{v}_{nt}^{(-j)}(v_j) | \boldsymbol{\beta}). \quad (2.3)$$

Step 2 Replacing $\lambda_j(v_{jit}) F_{jt}$ (or equivalently, $\gamma_{jt}(v_{jit})$) with $\hat{\gamma}_{jt}(v_{jit} | \boldsymbol{\beta}_0)$ in model (1.3) allows us to estimate $\boldsymbol{\beta}_0(\tau)$.

Specifically, we construct the local linear estimator by minimizing the following objective function:

$$\min_{\mathbf{a}, \mathbf{b}} \sum_{t=1}^T \|\mathcal{Y}_t - \mathcal{X}_t(\mathbf{a} + \mathbf{b}(\tau_t - \tau))\|^2 K\left(\frac{\tau_t - \tau}{h_0}\right), \quad (2.4)$$

where $\mathcal{Y}_t = (\hat{\gamma}_{1t}^*, \dots, \hat{\gamma}_{Nt}^*)^\top$ and $\mathcal{X}_t = (\hat{\mathbf{x}}_{1t}^*, \dots, \hat{\mathbf{x}}_{Nt}^*)^\top$ with $\hat{\gamma}_{it}^* = y_{it} - \frac{1}{N} \sum_{n=1}^N \sum_{j=1}^J \mathbf{s}_t(\mathbf{v}_{nt}^{(-j)}(v_{jit})) \mathbf{y}_t$ and $\hat{\mathbf{x}}_{it}^* = \mathbf{x}_{it} - \frac{1}{N} \sum_{n=1}^N \sum_{j=1}^J \mathbf{s}_t(\mathbf{v}_{nt}^{(-j)}(v_{jit})) \mathbf{x}_t$. Here, (2.4) gives

$$\hat{\boldsymbol{\beta}}(\tau) = [\mathbf{I}_p, \mathbf{0}_p] \left(\mathcal{X}^\top(\tau) \mathbf{W}(\tau) \mathcal{X}(\tau) \right)^{-1} \mathcal{X}^\top(\tau) \mathbf{W}(\tau) \mathcal{Y}, \quad (2.5)$$

where $\mathcal{Y} = (\mathcal{Y}_1^\top, \dots, \mathcal{Y}_T^\top)^\top$, and

$$\mathcal{X}(\tau) = \begin{pmatrix} \mathcal{X}_1^\top, \dots, \mathcal{X}_T^\top \\ (\frac{\tau_1 - \tau}{h_0}) \mathcal{X}_1^\top, \dots, (\frac{\tau_T - \tau}{h_0}) \mathcal{X}_T^\top \end{pmatrix}^\top.$$

Step 3 Substituting $\hat{\boldsymbol{\beta}}(\tau)$ into (2.3) yields an estimator for the product of factors and loadings: $\hat{\gamma}_{jt}(v_j | \hat{\boldsymbol{\beta}})$. Using the identification conditions stated in Assumption 1, we estimate the loadings by

$$\hat{\lambda}_j(v_j) = \frac{\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}(v_j | \hat{\boldsymbol{\beta}})}{\sqrt{\int \left(\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}(v_j | \hat{\boldsymbol{\beta}}) \right)^2 p_{v,j}(v_j) dv_j}}, \quad (2.6)$$

where $p_{v,j}(v_j)$ denotes the PDF of v_{jit} .¹ Finally, the factors are estimated by

$$\hat{F}_{jt} = \left(\sum_{i=1}^N \hat{\lambda}_j^2(v_{jit}) \right)^{-1} \sum_{i=1}^N \hat{\lambda}_j(v_{jit}) \hat{\gamma}_{jt}(v_{jit} | \hat{\boldsymbol{\beta}}). \quad (2.7)$$

In what follows, we derive the asymptotic distributions for the proposed estimators $\hat{\boldsymbol{\beta}}(\tau)$, $\hat{\lambda}_j(v_j)$, and $\hat{F}_{jt}$, respectively. Selecting appropriate bandwidths is essential to these estimations and is thoroughly discussed in Appendix A of the online supplement.

To proceed, we define some additional notations. Let $p_{v,j}(v_j)$, $p_v(\mathbf{v})$ and $p_v^{(-j)}(\mathbf{v}^{(-j)})$ be the probability density functions of v_{jit} , $\mathbf{v}_{it}$ and $\mathbf{v}_{it}^{(-j)}$, respectively. In addition, let $\phi_{it} = \sum_{j=1}^J p_v^{-1}(\mathbf{v}_{it}) p_{v,j}(v_{jit}) p_v^{(-j)}(\mathbf{v}_{it}^{(-j)})$, $\mathbf{m}_{x,t}(\mathbf{v}) = E[\mathbf{x}_{it} | \mathbf{v}_{it} = \mathbf{v}]$, $\mathbf{x}_{it}^* = \mathbf{x}_{it} - \sum_{j=1}^J E[\mathbf{m}_{x,t}(\mathbf{v}_{nt}^{(-j)}(v_j)) | v_j = v_{jit}]$, and $\mathbf{z}_{it} = \mathbf{x}_{it}^* - \phi_{it} E[\mathbf{x}_{it}^*]$. $\mathcal{F}_i^{(-\infty, 0)}$ and $\mathcal{F}_n^{(s, \infty)}$ denote the σ -fields generated by $\{\mathbf{x}_{it}, \mathbf{v}_{it}, \varepsilon_{it}; t \leq 0\}$ and $\{\mathbf{x}_{nt}, \mathbf{v}_{nt}, \varepsilon_{nt}; t \geq s\}$, respectively. The α -mixing coefficient $\alpha_{in}(s)$ is then defined as follows:

$$\alpha_{in}(s) = \sup_{\mathbf{A} \in \mathcal{F}_i^{(-\infty, 0)}, \mathbf{B} \in \mathcal{F}_n^{(s, \infty)}} |P(\mathbf{AB}) - P(\mathbf{A})P(\mathbf{B})|.$$

Assumption 2 (Weak dependence and kernel estimation).

- Suppose that $\{\mathbf{x}_t, \mathbf{v}_t, \boldsymbol{\varepsilon}_t\}$ are strictly stationary and α -mixing. The mixing coefficient $\alpha_{in}(s)$ satisfies $\sum_{i=1}^N \sum_{n=1}^N \sum_{s=1}^\infty (\alpha_{in}(s))^{\delta/(4+\delta)} = O(N)$ and $\sum_{i=1}^N \sum_{n=1}^N (\alpha_{in}(0))^{\delta/(4+\delta)} =$

¹When $p_{v,j}(v_j)$ is unknown, it can be replaced with its nonparametric estimate to construct the loading estimators. This issue will be discussed in detail at a later stage.

$O(N)$, where $\delta > 0$ is chosen such that $E[\|\mathbf{x}_{it}\|^{4+\delta}] < \infty$ and $E[\|\varepsilon_{it}\|^{4+\delta}] < \infty$.

2. Suppose that $\{\mathbf{x}_{it}, \mathbf{v}_{it}, \varepsilon_{it}\}$ are identically distributed for $i = 1, \dots, N$, with $E[\varepsilon_{11}] = 0$ and $\sigma_\varepsilon^2 = E[\varepsilon_{11}^2] > 0$.
3. $\{\varepsilon_{it}\}$ are independent of $\{\mathbf{v}_{js}, \mathbf{x}_{js}\}$ for all (i, j, t, s) .
4. $p_v(\mathbf{v})$ has continuous derivatives of up to the second order on the compact support of $\mathbf{v}_{it}$.
5. The unknown functions $\beta_0(\tau)$ and $\lambda_j(v_j)$ have continuous derivatives of up to the second order on $\tau \in [0, 1]$ and the support of v_{jit} , respectively.
6. The kernel function $K(\cdot)$ is Lipschitz continuous with compact support on $[-1, 1]$.
7. The bandwidth h_j satisfies that $h_j \rightarrow 0$, $\lim_{N,T \rightarrow \infty} NTh_j^5 < \infty$, $Nh_j \rightarrow \infty$ and $Th_j \rightarrow \infty$, for $j = 0, \dots, J$. In addition, we assume $Nh_0^4 \rightarrow 0$, $Nh_{\min}^8 \rightarrow 0$, $\frac{Th_0}{N} \rightarrow 0$ and $(Nh_j)^{(2+\delta^*/2)}h_{\min}^2/\log N \rightarrow \infty$, for some $\delta^* \in (0, \delta)$.

Assumption 2 introduces a set of standard conditions commonly used in the literature. **Assumptions 2.1–2** impose weak correlations across both dimensions, which is similar to Assumption 3.4 in Fan, Liao, and Wang (2016) and Assumption A1 in Connor, Hagmann, and Linton (2012). **Assumption 2.3** requires exogeneity. The conditions in **Assumptions 2.4–6** are standard in the local linear estimation literature (see, e.g., Chen, Gao, and Li 2012). **Assumption 2.7** imposes specific restrictions on the bandwidths.

Assumption 3 (Asymptotic covariance matrices).

1. Let $\Sigma_x = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N E[\mathbf{x}_{i1}^* \mathbf{x}_{i1}^{*\top}]$ be positive definite.
2. Let $\sigma_{in}^2(s) = E[\varepsilon_{i1} \varepsilon_{n,1+s}]$ and $\Sigma_{in}(s) = E[\mathbf{z}_{i1} \mathbf{z}_{n,1+s}^\top]$. Suppose that

$$\Sigma_\beta^0 = \lim_{N \rightarrow \infty} N^{-1} v_0 \sum_{i=1}^N \sum_{n=1}^N \left(\sum_{s=-\infty}^{\infty} \Sigma_{in}(s) \sigma_{in}^2(s) \right)$$

is positive definite, where $v_0 = \int K^2(v) dv$.

3. Let $\Sigma_{ins}^V(v_j, u_j) = \iint p_{v,ins}(\mathbf{v}, \mathbf{u}) \tilde{p}_{v,j}(\mathbf{v}) \tilde{p}_{v,j}(\mathbf{u}) d\mathbf{v}^{(-j)} d\mathbf{u}^{(-j)}$, where $p_{v,ins}(\cdot, \cdot)$ denotes the joint probability density function of $\mathbf{v}_{i1}$ and $\mathbf{v}_{n,1+s}$, and $\tilde{p}_{v,j}(\mathbf{v}) = p_v^{-1}(\mathbf{v}) p_v^{(-j)}(\mathbf{v}^{(-j)})$. In addition, let $\Sigma_j^V(v_j) = \lim_{N \rightarrow \infty} N^{-1} v_0 \sum_{i=1}^N \sum_{n=1}^N (\sum_{s=-\infty}^{\infty} \Sigma_{ins}^V(v_j, v_j) \sigma_{in}^2(s)) > 0$, for $j = 1, \dots, J$.
4. For any given j and t , suppose that $(Nh_j)^{-1/2} \sum_{i=1}^N K\left(\frac{v_{jit}-v_j}{h_j}\right) \tilde{p}_{v,j}(\mathbf{v}_{it}^{(-j)}(v_j)) \varepsilon_{it} \xrightarrow{D} \mathcal{N}\left(0, \Sigma_{jt}^V(v_j)\right)$, where $\Sigma_{jt}^V(v_j)$ is a positive scalar.
5. For any given j and t , suppose that $(N)^{-1/2} \sum_{i=1}^N \tilde{\xi}_{jit} \varepsilon_{it} \xrightarrow{D} \mathcal{N}\left(0, \Sigma_{jt}^F\right)$, where Σ_{jt}^F is a positive scalar and $\tilde{\xi}_{jit} = \frac{1}{Nh_j} \sum_{n=1}^N K\left(\frac{v_{jnt}-v_{jit}}{h_j}\right) \lambda_j(v_{jnt}) \tilde{p}_{v,j}(\mathbf{v}_{it}^{(-j)}(v_{jnt}))$.

In **Assumption 3**, we introduce asymptotic covariance matrices and assume their positive definiteness. In practice, these matrices can be estimated by their sample versions.

With these conditions in hand, the following theorem summarizes the asymptotic distributions of the proposed PMI estimators.

Theorem 1 (CLTs).

Let **Assumptions 1–3** hold. As $N, T \rightarrow \infty$,

1. for any given $\tau \in (0, 1)$,

$$\sqrt{NTh_0} (\hat{\beta}(\tau) - \beta_0(\tau) - \mathbf{a}(\tau)) \xrightarrow{D} \mathcal{N}(\mathbf{0}_p, \Sigma_\beta),$$

where $\Sigma_\beta = \Sigma_x^{-1} \Sigma_\beta^0 \Sigma_x^{-1}$ and the bias term $\mathbf{a}(\tau) = O_P(\sum_{j=0}^J h_j^2)$. Additionally, if $\sum_{j=1}^J h_j^2 = o(h_0^2)$, then $\mathbf{a}(\tau) = \frac{\mu_2 h_0^2}{2} \beta_0''(\tau)(1 + o_P(1))$, where $\beta_0''(\tau)$ denotes the second-order derivative of $\beta_0(\tau)$.

2. for any given j, t and an internal point v_j ,

$$\sqrt{Nh_j} (\hat{\gamma}_{jt}(v_j) \hat{\beta} - \gamma_{jt}(v_j)) \xrightarrow{D} \mathcal{N}(0, \Sigma_{jt}^V(v_j)),$$

where $\gamma_{jt}(v_j) = \lambda_j(v_j) F_{jt}$.

3. for any given j and an internal point v_j ,

$$\sqrt{NTh_j} (\hat{\lambda}_j(v_j) - \lambda_j(v_j) - c_j(v_j)) \xrightarrow{D} \mathcal{N}(0, \Sigma_j^V(v_j)),$$

where $\Sigma_j^V(v_j) = \bar{F}_j^{-2} \Sigma_j^V(v_j)$ and $c_j(v_j) = O_P(\sum_{j=0}^J h_j^2)$. Additionally, if $\sum_{k=0, \neq j}^J h_k^2 = o(h_j^2)$, then $c_j(v_j) = \frac{\mu_2 h_j^2}{2} \lambda_j''(v_j)(1 + o_P(1))$, where $\lambda_j''(v_j)$ denotes the second-order derivative of $\lambda_j(v_j)$.

4. for any given j and t , $\sqrt{N} (\hat{F}_{jt} - F_{jt}) \xrightarrow{D} \mathcal{N}(0, \Sigma_{jt}^F)$.

In **Theorem 1.1**, we establish a $\sqrt{NTh_0}$ consistency result for $\hat{\beta}(\tau)$, which achieves the optimal rate of convergence for nonparametric kernel estimation for panel data models. Under the condition: $\sum_{j=1}^J h_j^2 = o(h_0^2)$, the bias term simplifies to $\frac{\mu_2 h_0^2}{2} \beta_0''(\tau)$. This is consistent with conventional results in the nonparametric time-varying panel data literature (e.g., Theorem 3.2 of Chen, Gao, and Li (2012)). Without this condition, the leading order of the bias term becomes $O_P(h_0^2 + \sum_{j=1}^J h_j^2)$. **Theorem 1.2** demonstrates that the marginal integration estimator $\hat{\gamma}_{jt}(v_j) \hat{\beta}$ consistently recovers the product of loading functions and factors. Additionally, **Theorem 1.3** establishes the asymptotic distribution for loading estimators with the optimal rate of $\sqrt{NTh_j}$ under the α -mixing conditions in **Assumption 2**. The bias term is proportional to h_j^2 under the condition: $h_0^2 + \sum_{k \neq j} h_k^2 = o(h_j^2)$. Finally, **Theorem 1.4** shows that factor estimators achieve a parametric convergence rate of $\sqrt{N}$. These results align with the CLTs established for semiparametric factor models in Theorem 1 of Connor, Hagmann, and Linton (2012). It is noteworthy that the different convergence rates reflect the sequential order of estimation. We first construct an estimator for $\beta(\tau)$ by expressing $\gamma_{jt}(v_j)$ in terms of $\beta(\tau)$. The estimator of $\gamma_{jt}(v_j)$ is then obtained by substituting $\hat{\beta}(\tau)$. This sequential estimation leads to a faster convergence rate for $\hat{\beta}(\tau)$. A similar sequential process is applied when estimating the loadings and factors.

We can derive the expressions of the asymptotic covariance matrices $\Sigma_{jt}^V(v_j)$ and Σ_{jt}^F under additional restrictions on the weak cross-sectional dependence of the loading covariates and

random errors. For example, if $\mathbf{v}_{it}$ and ε_{it} are iid across i , we can use straightforward algebra to obtain

$$\Sigma_{jt}^{\gamma}(v_j) = \sigma_{\varepsilon}^2 \int \frac{p_v^{(-j)2}(\mathbf{v}^{(-j)})}{p_v(\mathbf{v})} d\mathbf{v}^{(-j)}. \quad (2.8)$$

This result is consistent with the conventional covariance matrix for the MI estimator (e.g., Theorem 3.2 of Gao, Lu, and Tjøstheim 2006). Similarly, we have

$$\Sigma_{jt}^F = \sigma_{\varepsilon}^2 \int_{\mathbf{v}} \lambda_j^2(v_j) \frac{(p_v^{(-j)}(\mathbf{v}^{(-j)})p_{vj}(v_j))^2}{p_v(\mathbf{v})} d\mathbf{v}, \quad (2.9)$$

under the iid condition. Additionally, if we impose a further assumption that v_{jit} and $\mathbf{v}_{it}^{(-j)}$ are independent, the expression of Σ_{jt}^F simplifies to $\Sigma_{jt}^F = \sigma_{\varepsilon}^2 \int_{v_j} \lambda_j^2(v_j) p_{vj}(v_j) dv_j = \sigma_{\varepsilon}^2$, where the second equality holds under the identification conditions stated in Assumption 1. In this simplified scenario, we obtain an asymptotic covariance matrix for $\hat{F}_{jt}$ that matches the OLS estimator when the loading functions are observable. This demonstrates that, under certain conditions, our estimator for the factors exhibits the same efficiency as the OLS estimator in a traditional regression setting.

In the construction of $\hat{\lambda}_j(v_j)$, knowledge of the PDF function ($p_{vj}(v_j)$) is required, which may not always be available. When $p_{vj}(v_j)$ is unknown, it can be replaced with its nonparametric estimate:

$$\hat{p}_{vj}(v_j) = \frac{1}{NTh_j} \sum_{i=1}^N \sum_{t=1}^T K\left(\frac{v_{jit} - v_j}{h_j}\right).$$

Using $\hat{p}_{vj}(v_j)$, the loadings estimators can be reconstructed as follows:

$$\tilde{\lambda}_j(v_j) = \frac{\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}(v_j | \hat{\beta})}{\sqrt{\int \left(\frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}(v_j | \hat{\beta})\right)^2 \hat{p}_{vj}(v_j) dv_j}}.$$

In the following corollary, we demonstrate that $\tilde{\lambda}_j(v_j)$ has an asymptotically equivalent distribution to $\hat{\lambda}_j(v_j)$.

Corollary 1.

Let Assumptions 1–3 hold. As $N, T \rightarrow \infty$, for any j and a given internal point v_j ,

$$\sqrt{NTh_j} (\tilde{\lambda}_j(v_j) - \lambda_j(v_j) - c_j(v_j)) \xrightarrow{D} \mathcal{N}(0, \Sigma_j^{\lambda}(v_j)),$$

where $\Sigma_j^{\lambda}(v_j)$ and $c_j(v_j)$ are defined in Theorem 1.

Corollary 1 demonstrates that replacing the unknown density function $p_{vj}(v_j)$ with its nonparametric estimators will not alter the asymptotic properties and we can obtain an identical CLT as $p_{vj}(v_j)$ is known. This result confirms the robustness of our method, indicating that accurate loadings estimation is feasible even when the distributions of loading covariates are not directly observed.

2.2. Testing for Parametric Specification

In this section, we consider a specification testing of the semi-parametric model (1.3). Specifically, we are interested in testing the null hypothesis of the following parametric specification:

$$\mathcal{H}_0 : \beta_{l0}(\tau) = \beta_l(\tau, \theta_{l0}) \text{ and}$$

$$\lambda_j(v_j) = \lambda_j(v_j, \phi_{j0}), \text{ for } 1 \leq l \leq p \text{ and } 1 \leq j \leq J,$$

where $\beta_l(\tau, \theta_{l0})$ is a parametric time-varying function with q_l -dimensional unknown parameter $\theta_{l0} \in \Theta_l$, and $\lambda_j(v_j, \phi_{j0})$ is a parametric loading function with r_j -dimensional unknown parameter $\phi_{j0} \in \Phi_j$ with Θ_l and Φ_j being the parameter spaces of $\mathbb{R}^{q_l}$ and $\mathbb{R}^{r_j}$, respectively. Various forms of parametric functions can be used in the null hypothesis. For example, we may test the time-variation in the regression coefficients and the linearity of loading functions in model (1.3) by setting

$$\beta_{l0}(\tau, \theta_{l0}) = \theta_{l0} \text{ and } \lambda_j(v_j, \phi_{j0}) = \phi_{j0} v_j.$$

For power analysis, we consider a sequence of alternatives:

$$\mathcal{H}_1 : \beta_{l0}(\tau) = \beta_l(\tau, \theta_{l0}) + v_{NT} \Delta_{\beta,l}(\tau) \text{ and}$$

$$\lambda_j(v_j) = \lambda_j(v_j, \phi_{j0}) + v_{NT} \Delta_{\lambda,j}(v_j),$$

for $1 \leq l \leq p$ and $1 \leq j \leq J$, where $\Delta_{\beta,l}(\tau)$ and $\Delta_{\lambda,j}(v_j)$ are nonparametrically unknown functions of τ and v_j satisfying $\int_0^1 \|\Delta_{\beta,l}(\tau)\|^2 d\tau < \infty$ and $E\|\Delta_{\lambda,j}(v_{jit})\|^2 < \infty$, respectively, and v_{NT} is a sequence of real numbers. For power analysis, we consider two cases for v_{NT} : the global alternative with $v_{NT} = 1$, and the local alternative with $v_{NT} = (NT)^{-1/2}(h_0 H_j)^{-1/4}$.

There exists an extensive literature on model specification testing. In this section we extend the relevant literature, such as Gao and Gijbels (2008) and Su, Jin, and Zhang (2015), to construct a test statistic based on the parametrically estimated residuals under the null hypothesis:

$$L_{NT,0} = \frac{1}{NT\sqrt{h_0 H_j}} \sum_{i,n=1}^N \sum_{t=1}^T \sum_{s=1, \neq t}^T K\left(\frac{\tau_t - \tau_s}{h_0}\right) \times \prod_{j=1}^J K\left(\frac{v_{jit} - v_{jns}}{h_j}\right) \hat{e}_{it} \hat{e}_{ns},$$

where $\{\hat{e}_{it}\}$ are residuals from estimating the restricted model. Under $\mathcal{H}_0$, the conventional parametric methods (e.g., OLS) can be used for the consistent estimation of the unknown parameters. A detailed discussion of these methods is omitted here due to page constraints. Let $\hat{\theta}_l$, $\hat{\phi}_j$ and $\hat{\mathbf{F}}_t$ be consistent parameter estimators of θ_{l0} , ϕ_{j0} , and $\mathbf{F}_t$, respectively, under $\mathcal{H}_0$. Also, let $\mathcal{K}_{in,ts} = \prod_{j=1}^J K\left(\frac{v_{jit} - v_{jns}}{h_j}\right)$ and $\bar{\Delta}_{e,it} = \sum_{l=1}^p x_{it}^{(l)} \Delta_{\beta,l}(\tau_t) + \sum_{j=1}^J \Delta_{\lambda,j}(v_{jit}) F_{jt}$. To investigate the asymptotic behavior of the test statistic under the null and alternative hypotheses, we need the following assumption.

Assumption 4.

1. Suppose that the parametric estimators $\hat{\theta}_l$, $\hat{\phi}_j$, and $\hat{\mathbf{F}}_t$ are $\sqrt{NT}$, $\sqrt{NT}$ and $\sqrt{N}$ -consistent estimators of θ_{l0} , ϕ_{j0} , and $\mathbf{F}_t$, respectively under $\mathcal{H}_0$.

2. The null parametric functions $\beta_{0l}(\tau, \theta_{l0})$ and $\lambda_j(v_j, \phi_{j0})$ are Lipschitz continuous with respect to θ_{l0} and ϕ_{j0} , respectively.
3. For each i , $E[\varepsilon_{it}|\mathbb{F}_{t-1}] = 0$, $\sigma_\varepsilon^2 = E[\varepsilon_{it}^2|\mathbb{F}_{t-1}]$, and $\sigma_{\varepsilon, in}^2 = E[\varepsilon_{it}\varepsilon_{nt}|\mathbb{F}_{t-1}]$, where $\mathbb{F}_t$ is the sigma-field generated by $\{\varepsilon_t, \varepsilon_{t-1}, \dots\}$.
4. Under $\mathcal{H}_1$, $(N^2 T^2 h_0 H_J)^{-1} \sum_{i,n=1}^N \sum_{t,s=1}^T K\left(\frac{\tau_t - \tau_s}{h_0}\right) \mathcal{K}_{in,ts} \overline{\Delta}_{e,ns} \overline{\Delta}_{e,it} \xrightarrow{P} \overline{\Delta}_L$, where $\overline{\Delta}_L$ is a positive and finite number.
5. The bandwidth h_0 and H_J satisfy that $NT(h_0 H_J)^{1/2} \rightarrow \infty$.

In [Assumption 4](#), we introduce some conventional conditions to establish the asymptotic distribution of the test statistic L_{NT} . [Assumption 4.1](#) requires consistency of the parametric estimators under $\mathcal{H}_0$. This condition is mild and can be satisfied by most conventional parametric estimators for different null hypotheses. It extends [Assumption A.2\(a\)](#) of Gao and Gijbels (2008) for the specification test of panel data models and is similar to the condition used in [Assumption A.5](#) of Su and Lu (2013). [Assumption 4.2](#) requires the Lipschitz continuity of the parametric function in null hypothesis. With [Assumption 4.3](#), we can apply the CLT for U -statistic to study the asymptotic distribution of L_{NT} . [Assumption 4.4](#) regulates the asymptotic behavior of the test statistic under the local alternative with $v_{NT} = (NT)^{-1/2}(h_0 H_J)^{-1/4}$. [Assumption 4.5](#) imposes further restrictions on the bandwidths, which can be satisfied by many conventional bandwidth choices. For example, if $h_j = O((NT)^{-1/5})$ and $J = 4$, this condition holds, as $N^2 T^2 h_0 H_J = O(NT)$.

With [Assumption 4](#), we can show that the asymptotic covariance of L_{NT} is given by $\sigma_L^2 = 2v_0^{J+1} \sigma_\varepsilon^4 \int p_v^2(\mathbf{v}) d\mathbf{v}$ under the null parametric hypothesis. We put its derivation in the proof of [Theorem 2](#) in the online supplement. Then, we construct a consistent estimator for σ_L^2 as follows: $\widehat{\sigma}_L^2 = 2v_0^{J+1} \widehat{\sigma}_\varepsilon^4 \int \widehat{p}_v^2(\mathbf{v}) d\mathbf{v}$, where $\widehat{\sigma}_\varepsilon^2 = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \widehat{e}_{it}^2$ and $\widehat{p}_v(\mathbf{v}) = (NTH_J)^{-1} \sum_{i=1}^N \sum_{t=1}^T \mathcal{K}_{it}(\mathbf{v})$. With $\widehat{\sigma}_L^2$, we finally introduce the standardized test statistic as follows:

$$L_{NT} = \frac{1}{\sqrt{\widehat{\sigma}_L^2}} L_{NT,0}.$$

We summarize the asymptotic properties of L_{NT} under $\mathcal{H}_0$ and $\mathcal{H}_1$ in the following theorem.

Theorem 2 (Asymptotic properties for L_{NT}).

Let [Assumptions 1, 2, and 4](#) hold. As $N, T \rightarrow \infty$,

1. $L_{NT} \xrightarrow{D} \mathcal{N}(0, 1)$, under $\mathcal{H}_0$;
2. $L_{NT} \xrightarrow{D} \mathcal{N}(\overline{\Delta}_L / \sigma_L, 1)$, under $\mathcal{H}_1$ with $v_{NT} = (NT)^{-1/2}(h_0 H_J)^{-1/4}$, where $\overline{\Delta}_L$ is defined in [Assumption 4](#);
3. $P(L_{NT} > L_\alpha) \rightarrow 1$, under $\mathcal{H}_1$ with $v_{NT} = 1$, where L_α is the α -level critical value of $\mathcal{N}(0, 1)$.

[Theorem 2](#) establishes the asymptotic properties of the test statistic under three different hypotheses: the null, the local alternative, and the global alternative. Under the null hypothesis, the test statistic converges to a standard normal distribution, ensuring that the test has the correct asymptotic size. When considering the local alternative hypothesis with $v_{NT} =$

$(NT)^{-1/2}(h_0 H_J)^{-1/4}$, the power of the test is determined by $\overline{\Delta}_L$, which characterizes the deviations from the null hypothesis. [Theorem 23](#) further indicates that, under the global alternative hypothesis, the power of the test statistic can converge to one, confirming the consistency of the test. By combining the results from [Theorem 2.2](#) and 3, we can also conclude that the local power will converge to one if v_{NT} converges at a slower rate than $(NT)^{-1/2}(h_0 H_J)^{-1/4}$. These results demonstrate the usefulness of the test in detecting both local and global departures from the null hypothesis.

2.3. An Extension of the Semiparametric Model

A potential limitation of model (1.3) is its requirement that the number of factors equals the number of observable covariates. To accommodate a more flexible relationship between the loadings and the factors, we relax this assumption. Specifically, we consider the following generalized model:

$$y_{it} = \mathbf{x}_{it}^\top \boldsymbol{\beta}_0(\tau_t) + \sum_{k=1}^K \bar{\lambda}_k(\mathbf{v}_{it}) F_{kt} + \varepsilon_{it}, \quad (2.10)$$

where K is number of factors and the k th loading $\bar{\lambda}_k(\mathbf{v}_{it}) = \sum_{j=1}^J \lambda_{kj}(v_{jit})$ is unknown function of $J \times 1$ vector of observable covariates $\mathbf{v}_{it}$ in an additive form. We assume K is known. Model (2.10) extends model (1.3) in the sense that it allows a different number of factors (K) from the number of loading covariates (J). This setting is similar to the factor structures considered by Chen, Roussanov, and Wang (2021) and Chen et al. (2023).

The PMI approach can be modified to construct consistent estimators for model (2.10) as follows:

Step 1. For any given $\boldsymbol{\beta}(\tau_t)$, recall that $\gamma_t(\mathbf{v} | \boldsymbol{\beta}) := E[y_{it} - \mathbf{x}_{it}^\top \boldsymbol{\beta}(\tau_t) | \mathbf{v}_{it} = \mathbf{v}]$. Under model (2.10), we have

$$\gamma_t(\mathbf{v} | \boldsymbol{\beta}_0) = \sum_{k=1}^K \sum_{j=1}^J \lambda_{kj}(v_j) F_{kt} := \sum_{j=1}^J \bar{\gamma}_{jt}(v_j), \quad (2.11)$$

where $\bar{\gamma}_{jt}(v_j) = \sum_{k=1}^K \lambda_{kj}(v_j) F_{kt}$. Drawing upon (2.3) and (2.11), we can still introduce the MI estimator (given $\boldsymbol{\beta}(\tau_t)$) for $\bar{\gamma}_{jt}(v_j)$: $\widehat{\gamma}_{jt}(v_j | \boldsymbol{\beta}) = \frac{1}{N} \sum_{i=1}^N \widehat{\gamma}_{it}(\mathbf{v}_{it}^{(-j)}(v_j) | \boldsymbol{\beta})$, where $\widehat{\gamma}_{it}(\mathbf{v} | \boldsymbol{\beta})$ is defined in (2.2).

Step 2. For each j , replace $\sum_{k=1}^K \lambda_{kj}(v_{jit}) F_{kt}$ (or equivalently, $\bar{\gamma}_{jt}(v_{jit})$) with $\widehat{\gamma}_{jt}(v_{jit} | \boldsymbol{\beta}_0)$ in model (2.10) to estimate $\boldsymbol{\beta}_0(\tau)$ using the local linear method as in the Step 2 of the PMI estimation described in [Section 2.1](#). Denote $\widehat{\boldsymbol{\beta}}(\tau)$ as the estimator.

Step 3. Construct an estimator for $\bar{\gamma}_{jt}(v_j)$ by plugging $\widehat{\boldsymbol{\beta}}(\tau)$ into $\widehat{\gamma}_{jt}(v_j | \boldsymbol{\beta})$: $\widehat{\gamma}_{jt}(v_j) = \widehat{\gamma}_{jt}(v_j | \widehat{\boldsymbol{\beta}})$. Let $\dot{\gamma}_{jit} = \widehat{\gamma}_{jt}(v_{jit})$ and $\dot{\mathbf{r}}_i = (\sum_{j=1}^J \dot{\gamma}_{ji1}, \dots, \sum_{j=1}^J \dot{\gamma}_{jiT})^\top$. The factors can be recovered through the principal component analysis:

$$\frac{1}{NT} \sum_{i=1}^N \dot{\mathbf{r}}_i \dot{\mathbf{r}}_i^\top \dot{\mathbf{F}} = \dot{\mathbf{F}} \dot{\mathbf{V}}_{NT}, \quad (2.12)$$

where $\dot{\mathbf{V}}_{NT}$ is a $K \times K$ diagonal matrix containing the largest K eigenvalues of $\frac{1}{NT} \sum_{i=1}^N \dot{\mathbf{r}}_i \dot{\mathbf{r}}_i^\top$ in descending order, and $\dot{\mathbf{F}}$ is

a $T \times K$ matrix of the corresponding K eigenvectors satisfying that $\frac{1}{T}\tilde{\mathbf{F}}^\top \tilde{\mathbf{F}} = \mathbf{I}_K$.

Step 4. Let $\lambda_j(v_j) = (\lambda_{1j}(v_j), \dots, \lambda_{Kj}(v_j))^\top$. For each j and $v_j \in [\underline{v}_j, \bar{v}_j]$, obtain the local linear estimators by minimizing the following objective function:

$$\min_{\mathbf{a}, \mathbf{b}} \sum_{i=1}^N \sum_{t=1}^T (\tilde{\Gamma}_{jit} - \tilde{\mathbf{F}}_t^\top \mathbf{a} - (v_{jit} - v_j) \tilde{\mathbf{F}}_t^\top \mathbf{b})^2 K\left(\frac{v_{jit} - v_j}{h_j}\right), \quad (2.13)$$

where $\tilde{\Gamma}_{jit} = \hat{\gamma}_{jt}(v_{jit})$ and $\tilde{\mathbf{F}}_t$ is the t th column of $\tilde{\mathbf{F}}^\top$.

Let $\mathcal{F}_{jit}(v_j) = (\tilde{\mathbf{F}}_t^\top, (\frac{v_{jit} - v_j}{h_j}) \tilde{\mathbf{F}}_t^\top)^\top$, $\dot{\mathcal{F}}_j(v_j) = (\mathcal{F}_{j11}(v_j), \mathcal{F}_{j21}(v_j), \dots, \mathcal{F}_{jNT}(v_j))^\top$, $\tilde{\mathbf{F}}_j = (\tilde{\Gamma}_{j11}, \tilde{\Gamma}_{j21}, \dots, \tilde{\Gamma}_{jNT})^\top$, and $\dot{\mathbf{W}}_j(v_j) = \text{diag}\left(K\left(\frac{v_{j11} - v_j}{h_j}\right), K\left(\frac{v_{j21} - v_j}{h_j}\right), \dots, K\left(\frac{v_{jNT} - v_j}{h_j}\right)\right)$. It follows (2.13) that

$$\dot{\lambda}_j(v_j) = [\mathbf{I}_K, \mathbf{0}_K] \left(\dot{\mathcal{F}}_j^\top(v_j) \dot{\mathbf{W}}_j(v_j) \dot{\mathcal{F}}_j(v_j) \right)^{-1} \dot{\mathcal{F}}_j^\top(v_j) \dot{\mathbf{W}}_j(v_j) \tilde{\mathbf{F}}_j.$$

Finally, we standardize $\dot{\lambda}_j(v_j)$ to ensure that the identification conditions can be satisfied. Specifically, for $\lambda_{kj}(v_j)$,

$$\bar{\lambda}_{kj}(v_j) = \frac{\dot{\lambda}_{kj}(v_j)}{\sqrt{\int \dot{\lambda}_{kj}(v_j)^2 p_{v_j}(v_j) dv_j}},$$

where $\dot{\lambda}_{kj}(v_j)$ is the k th element of $\dot{\lambda}_j(v_j)$.

As evident from this estimation procedure, the PMI estimation can be applied to estimate a generalization of model (1.3) to accommodate different numbers of factors and loading covariates after necessary modifications. Using arguments similar to those in Theorem 1, we can establish the CLTs for the modified PMI estimators. Additionally, an alternative approach to estimate the factors and loading functions from $\tilde{\Gamma}_{jit}$ is the projected PCA method considered by Fan, Liao, and Wang (2016). Compared with the traditional PCA method, this approach can produce more accurate estimates by incorporating observable covariate information into the estimation of factors. Due to the page constraints, we leave these aspects for future research.

Up to this point, we have finished our theoretical investigation on (1.3). In the following sections, we examine the finite-sample performance of the proposed models and estimation methods using empirical and simulation examples.

3. Simulation Studies

In this section, we conduct several simulation experiments to evaluate the performance of our proposed estimators and test statistics.

3.1. On the Estimation Method

Examples 3.1 and 3.2 are designed to assess the effectiveness of the PMI estimation method.

Example 3.1. We consider the following data generating process (DGP):

$$y_{it} = x_{1,it} \beta_{10}(\tau_t) + x_{2,it} \beta_{20}(\tau_t) + \lambda_1(v_{1,it}) F_{1t} + \lambda_2(v_{2,it}) F_{2t} + \varepsilon_{it},$$

where $(\beta_{10}(\tau), \beta_{20}(\tau)) = (\sin(\frac{\pi\tau}{4}), \cos(\frac{\pi\tau}{4}))$, $(x_{1,it}, x_{2,it}) = (\eta_{1,it}, \eta_{2,it})$, $(\lambda_1(v_1), \lambda_2(v_2)) = (\sqrt{3}v_1, \sqrt{7}v_2^3)$, $(F_{1t}, F_{2t}) = (\tau_t, 1 + \tau_t^2)$, $(v_{1,it}, v_{2,it})^\top$ are independent and uniformly distributed on $[-1, 1]$, $(\eta_{1,1t}, \dots, \eta_{1,Nt})^\top$, $(\eta_{2,1t}, \dots, \eta_{2,Nt})^\top$ and $(\varepsilon_{1t}, \varepsilon_{2t}, \dots, \varepsilon_{Nt})^\top$ are independently generated as N -dimensional vectors of multivariate Student's t -distribution variables with 10 degrees of freedom, zero mean and a covariance matrix with $\sigma_{ij} = 0.5^{|i-j|}$ as its (i, j) th element.

Example 3.2. We consider the case where

$$x_{1,it} = \sqrt{\lambda_1(v_{1,it})^2 + \lambda_1(v_{2,it})^2} + \eta_{1,it} \quad \text{and} \\ x_{2,it} = \lambda_1(v_{1,it})^2 + \lambda_1(v_{2,it})^2 + \eta_{2,it}.$$

All other variables are generated in the same way as in Example 3.1.

We consider four additional types of time-varying coefficient functions in the online supplement. In each estimation, Epanechnikov kernel function $K(\mu) = \frac{3}{4}(1 - \mu^2)I(|\mu| \leq 1)$ is used in the local linear estimation of $\beta_0(\tau)$ and $\lambda_j(v_j)$. The bandwidth is selected using the conventional leave-one-unit-out cross-validation method as described in the online supplement.

We replicate the experiments 1000 times ($R = 1000$) and calculate the root-average of mean squared errors (RMSEs) for $(\beta_{10}(\tau), \beta_{20}(\tau))$, $(\lambda_1(v_1), \lambda_2(v_2))$, and (F_{1t}, F_{2t}) , respectively:

$$\text{RMSE}_\beta = \sqrt{\frac{1}{R} \sum_{r=1}^R \frac{1}{T} \sum_{t=1}^T \sum_{l=1}^2 \left| \hat{\beta}_l^{(r)}(\tau_t) - \beta_{l0}(\tau_t) \right|^2}, \\ \text{RMSE}_\lambda = \sqrt{\frac{1}{R} \sum_{r=1}^R \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \sum_{j=1}^2 \left| \hat{\lambda}_j^{(r)}(v_{jit}) - \lambda_j(v_{jit}) \right|^2}, \\ \text{RMSE}_F = \sqrt{\frac{1}{R} \sum_{r=1}^R \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^2 \left| \hat{F}_{jt}^{(r)} - F_{jt} \right|^2}.$$

The RMSEs for Examples 3.1 and 3.2 are provided in Table 1, which indicates that the RMSEs of $\hat{\beta}(\tau)$ decrease significantly as (N, T) increase from (50, 50) to (200, 200) in each simulation example. This finding is consistent with Theorem 1, which implies $\hat{\beta}(\tau)$ has $\sqrt{NTh_0}$ consistency. The simulation results from Example 3.2 also show the convergence of $\hat{\beta}(\tau)$ when the correlation between the explanatory variables and loadings is present. It verifies our arguments that the consistency of the proposed PMI method is not impaired by the dependency between $\mathbf{x}_{it}$ and $\mathbf{v}_{it}$. The RMSEs for the loading and factor estimates decrease as N increases. An increase in T does not eliminate estimation biases for the factor estimate, which is consistent with our theory in Theorem 1.3.

Table 1. Simulation results for RMSEs in Examples 3.1 and 3.2.

	N/T	Example 3.1				Example 3.2			
		50	100	150	200	50	100	150	200
RMSE $_{\beta}$	50	0.083	0.056	0.037	0.033	0.040	0.026	0.024	0.023
	100	0.030	0.028	0.022	0.020	0.026	0.022	0.019	0.017
	150	0.026	0.019	0.018	0.015	0.023	0.020	0.019	0.016
	200	0.023	0.018	0.016	0.014	0.021	0.018	0.017	0.015
RMSE $_{\lambda}$	50	0.297	0.222	0.210	0.183	0.224	0.206	0.194	0.186
	100	0.155	0.134	0.122	0.111	0.204	0.185	0.172	0.163
	150	0.140	0.121	0.110	0.103	0.192	0.171	0.159	0.150
	200	0.133	0.113	0.101	0.095	0.183	0.163	0.150	0.140
RMSE $_F$	50	0.440	0.347	0.347	0.198	0.190	0.194	0.192	0.192
	100	0.129	0.134	0.136	0.172	0.131	0.127	0.129	0.131
	150	0.107	0.108	0.110	0.110	0.106	0.106	0.105	0.107
	200	0.095	0.094	0.097	0.096	0.092	0.092	0.093	0.094

3.2. On the Specification Test

Examples 3.3 and 3.4 are used to evaluate the simulated sizes and powers of the proposed specification test.

Example 3.3. We consider the following DGP to compute the size of the misspecification test:

$$y_{it} = x_{1,it}\beta_{10} + x_{2,it}\beta_{20} + \lambda_{10}v_{1,it}F_{1t} + \lambda_{20}v_{2,it}F_{2t} + \varepsilon_{it},$$

where $(\beta_{10}, \beta_{20}) = (1, 1)$, $(\lambda_{10}, \lambda_{20}) = (1, 1)$, $(F_{1t}, F_{2t}) = (\tau_t^2, \tau_t)$ and the other variables are generated in the same way as in Example 3.2.

Example 3.4. We consider the following DGP to compute the power of the misspecification test:

$$y_{it} = x_{1,it}\beta_{10}(\tau_t) + x_{2,it}\beta_{20}(\tau_t) + \lambda_1(v_{1,it})F_{1t} + \lambda_2(v_{2,it})F_{2t} + \varepsilon_{it},$$

where $(\beta_{10}(\tau), \beta_{20}(\tau)) = (\beta_{10} + v_{NT} \sin(\frac{\pi\tau}{4}), \beta_{20} + v_{NT}\tau)$ and $(\lambda_1(v), \lambda_2(v)) = (\lambda_{10}v + v_{NT}v^3, \lambda_{20}v + v_{NT} \sin(\pi v))$. We consider three choices of v_{NT} to calculate the global and local powers, respectively: (a) $v_{NT} = 1$, (b) $v_{NT} = (NT)^{-1/2}(h_0H_J)^{-1/4}$ and (c) $v_{NT} = (NT)^{-1/3}(h_0H_J)^{-1/6}$. The other variables are generated in the same way as in Example 3.3. For Examples 3.3 and 3.4, we test the null hypothesis of the parametric model:

$$\mathcal{H}_0 : \beta_{10}(\tau) = \beta_{10}, \quad \beta_{20}(\tau) = \beta_{20}, \\ \lambda_1(v_1) = \lambda_{10}v_1, \quad \lambda_2(v_2) = \lambda_{20}v_2,$$

and treat $\beta_{10}, \beta_{20}, \lambda_{10}$, and λ_{20} as unknown parameters.

The number of bootstraps in the testing procedure is selected to be 500 and we repeat the test for 1000 times to compute the rejection rates under confidence levels 1%, 5%, and 10%, respectively. The rejection rates for Examples 3.3 and 3.4 are reported in Table 2.

Since the null hypothesis is true for Examples 3.3, the rejection rates for this example are the simulated sizes. As can be seen from Table 2, the proposed test can behave reasonably well in the sense that the simulated sizes are close to the corresponding significance levels across all sample sizes. In Examples 3.4, the parametric null hypothesis is false and therefore the rejection rates are estimated powers. The global powers ($v_{NT} =$

1) are all ones even when the sample sizes are small ($N = T = 50$, for example). Under the local alternative with $v_{NT} = (NT)^{-1/2}(h_0H_J)^{-1/4}$, the test power fails to converge to 1, which is in line with the theoretical development in Theorem 2.2. In terms of the local powers with $v_{NT} = (NT)^{-1/3}(h_0H_J)^{-1/6}$, they generally increase as either N or T increases and are approaching one when N and T are sufficiently large.

4. Empirical Application

In this section, we conduct an empirical analysis to evaluate the finite-sample performance of the proposed PMI estimation method. A substantial body of research has sought to explain differences in fund returns, as discussed in the finance literature (e.g., Jensen 1968; Ferson and Schadt 1996; Mamaysky, Spiegel, and Zhang 2008). In this section, we follow Blake et al. (2014) to study the relationship between mutual fund performance and three key fund characteristic variables: fund size, fund family size and management fees. Specifically, we aim to answer the following questions: (a) Do the funds that are managing smaller assets produce higher returns? (b) Do larger fund management companies outperform smaller ones? (c) Is higher management charge associated with higher fund performance? (d) Are the results for (a)–(c) robust across different time periods?

4.1. Variables and Data

We examine three fund-specific variables: relative size of assets among funds under management (rsa_{it}), relative size of fund family assets ($rsfa_{it}$) and average management charges (amc_{it}). As suggested by Blake et al. (2014), we need to control the effects of common risk factors in the market, which have been proved to play a significant role in explaining the difference in the fund performance (see, e.g., Fama and French 2010). A number of empirical studies have revealed the evidence of nonlinearity in the relationship between firm characteristics and security returns (e.g., Gu, Kelly, and Xiu 2020; Freyberger, Neuhierl, and Weber 2020). Additionally, Falkenstein (1996) documents the nonlinear preferences of U.S. mutual fund portfolio holdings for stock characteristics, such as size and liquidity, in quadratic forms. Building on this body of literature, we include four characteristics: size ($size_{it}$), value ($value_{it}$), momentum (mom_{it}), and volatility (vol_{it}) in the nonlinear loading functions, which are

Table 2. Rejection rates for Examples 3.3 and 3.4.

	N/T	$\alpha = 0.01$				$\alpha = 0.05$				$\alpha = 0.1$			
		50	100	150	200	50	100	150	200	50	100	150	200
Example 3.3	50	0.020	0.019	0.017	0.017	0.063	0.056	0.060	0.058	0.118	0.113	0.115	0.113
	100	0.014	0.015	0.012	0.017	0.060	0.063	0.054	0.060	0.115	0.112	0.115	0.109
	150	0.015	0.016	0.013	0.014	0.061	0.060	0.050	0.052	0.106	0.108	0.107	0.102
	200	0.016	0.014	0.011	0.011	0.055	0.058	0.051	0.054	0.094	0.105	0.096	0.103
Example 3.4 ($v_{NT} = 1$)	50	1	1	1	1	1	1	1	1	1	1	1	1
	100	1	1	1	1	1	1	1	1	1	1	1	1
	150	1	1	1	1	1	1	1	1	1	1	1	1
	200	1	1	1	1	1	1	1	1	1	1	1	1
Example 3.4 ($v_{NT} = (NT)^{-1/2}(h_0H_J)^{-1/4}$)	50	0.133	0.174	0.212	0.164	0.217	0.311	0.270	0.300	0.262	0.360	0.324	0.359
	100	0.136	0.163	0.227	0.213	0.204	0.296	0.335	0.337	0.302	0.366	0.389	0.421
	150	0.151	0.220	0.240	0.212	0.235	0.317	0.338	0.286	0.281	0.395	0.392	0.354
	200	0.201	0.215	0.244	0.220	0.288	0.329	0.365	0.351	0.375	0.407	0.450	0.447
Example 3.4 ($v_{NT} = (NT)^{-1/3}(h_0H_J)^{-1/6}$)	50	0.664	0.765	0.791	0.894	0.774	0.845	0.853	0.926	0.796	0.873	0.897	0.930
	100	0.779	0.827	0.913	0.951	0.845	0.907	0.950	0.979	0.891	0.939	0.975	0.992
	150	0.822	0.920	0.960	0.992	0.881	0.965	0.975	0.994	0.916	0.990	0.986	0.996
	200	0.882	0.947	1	1	0.928	0.981	1	1	0.944	0.995	1	1

computed based on fund portfolio holdings, to reflect the exposure of mutual funds to the common risk factors. In summary, we consider the following model for mutual fund excess returns (er_{it}):

$$er_{it} = \mathbf{x}_{it}\beta_0(\tau_t) + \lambda(\mathbf{v}_{it})^\top \mathbf{F}_t + \varepsilon_{it}, \quad (4.1)$$

where $\mathbf{x}_{it} = (\text{rsa}_{it}, \text{rsfa}_{it}, \text{amc}_{it})^\top$ and $\mathbf{v}_{it} = (\text{size}_{it}, \text{value}_{it}, \text{mom}_{it}, \text{vol}_{it})^\top$.²

We obtain monthly data of U.S. mutual funds from the Center for Research in Security Prices (CRSP) Survivor-Bias-Free Mutual Fund database from January 2008 to December 2019 period ($T = 143$). After excluding funds with incomplete records, initial total net assets (TNAs) under 10 million dollars, or less than 85% of holdings invested in equity markets, we finally have a sample of 177 mutual funds ($N = 177$).

To calculate fund returns, we compute monthly returns in excess of risk-free rates (er_{it}), using risk-free rate data are also collected from CRSP. Following Blake et al. (2014), we construct rsa_{it} , rsfa_{it} , and amc_{it} : rsa_{it} is the ratio of the fund's TNAs under management to the average value of TNAs across all funds in the previous month; rsfa_{it} is defined as the ratio of fund family's TNAs under management to the average value of all fund family TNAs in the previous month; and amc_{it} is the average management charges of all funds under management.

We calculate portfolio characteristics as weighted averages of the characteristics of stocks within the fund portfolio holdings, using the percentage of stock market capitalization in the portfolio as the weighting factor. Following the methodologies of Connor, Hagmann, and Linton (2012) and Fan, Liao, and Wang (2016), we compute the stock characteristics of size, value, momentum, and volatility as follows: the raw size characteristic for each stock in fund portfolio is the log value of the stock market capitalization in the previous month. The raw value

Table 3. Descriptive statistics and unit root test results.

Variables	Mean	SD	Z_{tbar}	p-value
er	0.003	0.048	-137.89	<0.01
size	6.200	1.396	-10.69	<0.01
value	-2.956	1.808	-8.23	<0.01
mom	0.048	0.125	-38.07	<0.01
vol	0.042	0.020	-26.76	<0.01
rsa	0.000	0.048	-119.36	<0.01
rsfa	0.000	0.038	-131.97	<0.01
amc	0.701	0.132	-12.47	<0.01

characteristic value equals the log ratio of the stock market capitalization to the book value in the previous month. The raw momentum characteristic is derived as the cumulative stock returns from the previous 6 months. The raw volatility is finally computed as the standard deviation of the stock returns in the previous 6 months. The data of stock market capitalization and book value are obtained from Compustat.

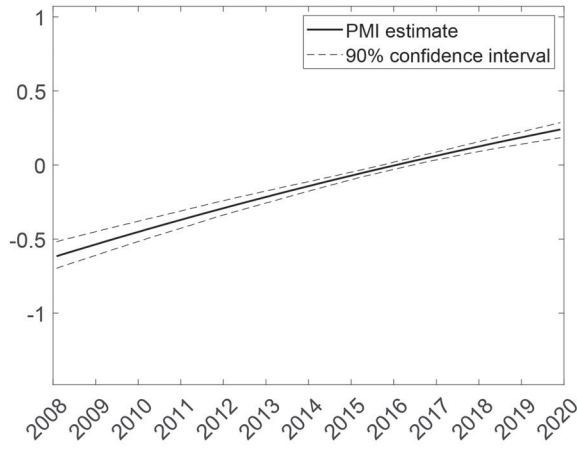
To assess the validity of model (1.3) for this dataset, we conduct unit root tests on both the fund characteristic variables and the fund-specific explanatory variables. We employ the well-known panel data IPS test proposed by Im, Pesaran, and Shin (2003) to detect potential unit root behavior in these variables. Table 3 reports the IPS test statistic Z_{tbar} , as defined in eq. (4.6) of Im, Pesaran, and Shin (2003). As indicated in Table 3, the results do not support the null hypothesis of a panel unit root for these variables, thereby confirming the validity of the stationarity assumption. Means and standard deviations of these variables are also provided in Table 3. Additionally, Figures B.1 and B.2 in the online supplement show plots of the raw data for these variables. For estimation, the four portfolio characteristics are standardized to the interval $[-1, 1]$.

4.2. Results from Estimation and Testing

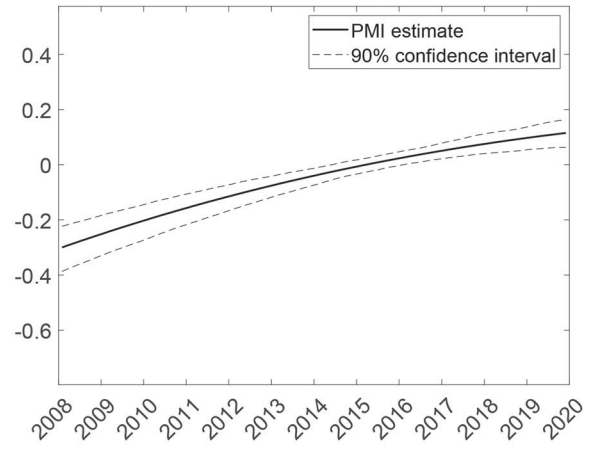
Before proceeding further, we conduct a sequence of specification tests to justify our setup in (4.1). Specifically, we conduct three hypothesis testing: (i) (model significance test) $\mathcal{H}_0^{(1)}: \beta_{k0}(\tau) = 0$ and $\lambda_j(v_j) = 0$, for $1 \leq k, j \leq 4$; (ii) (constancy test

²The economic meanings of rsa_{it} , rsfa_{it} and size_{it} are different, though they are all related to the fund size. rsa_{it} and rsfa_{it} reflect the size of assets under management of the fund and fund family, while size_{it} is the exposure of fund portfolio to the common size risk factor in market.

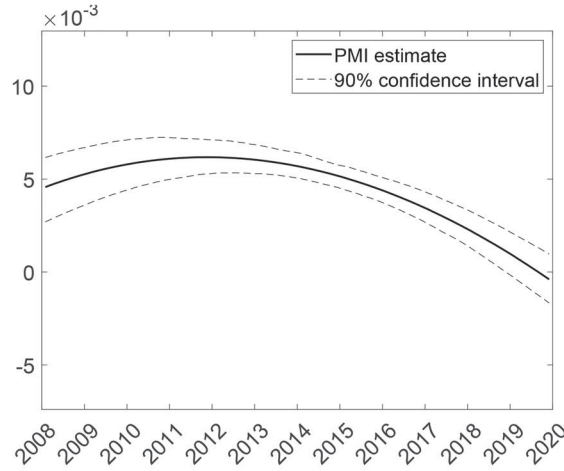
Initially, we included a time-varying intercept in our model, but the estimation results indicated that the intercept was not significant. Consequently, we remove the intercept from the model.



(a) rsa



(b) rsfa



(c) amc

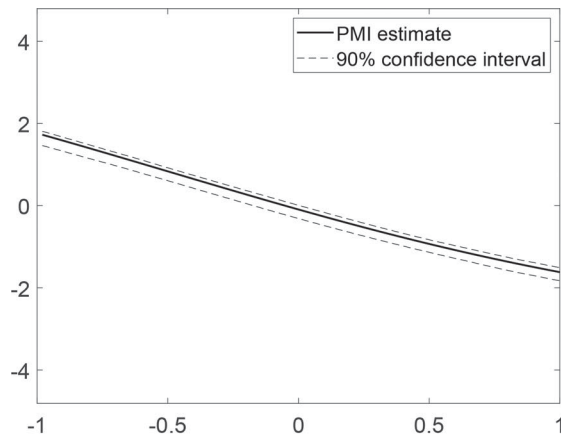
Figure 1. Estimated time-varying regression coefficient functions for model (4.1).

on the regression) $\mathcal{H}_0^{(2)}: \beta_{k0}(\tau) = \beta_{k0}$ and $\lambda_j(v_j) = 0$, where β_{k0} is an unknown parameter, for $1 \leq k, j \leq 4$, (iii) (test on the linearity) $\mathcal{H}_0^{(3)}: \beta_k(\tau) = \beta_{k0} + \beta_{k1}\tau$ and $\lambda_j(v_j) = \lambda_{j1}v_j$, where β_{k0}, β_{k1} and λ_{j1} are unknown parameters, for $1 \leq k, j \leq 4$. We obtain the values of the test statistic as 88.14, 43.09, and 24.33 for hypotheses (i), (ii), and (iii), respectively, with the corresponding bootstrap 0.01-level critical values of 5.61, 3.94, and 1.62. Therefore, we can conclude that model (4.1) is significant. In addition, these test results confirm the importance to document the time-variation in regression coefficients and the nonlinearity in loading functions.

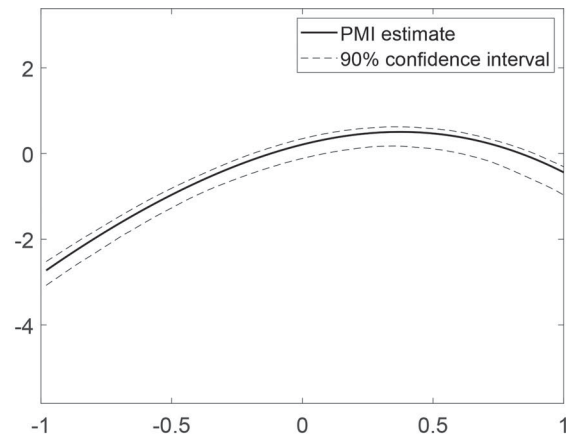
We now present the estimated coefficients in Figure 1. As shown, the estimated coefficient function for rsa_{it} is generally negative from 2008 to 2017 with an upward trend over this period. This suggests that, during this period, U.S. mutual fund returns were generally negatively related to the relative size of fund assets under management. This finding aligns with Blake et al.'s (2014) study on U.K. mutual funds. After 2017, however, this effect becomes positive. Similarly, the effects of fund fam-

ily size follow a comparable pattern: U.S. mutual fund returns are generally negatively associated with the size of fund family assets, but this negative impact on fund performance has been diminishing in recent years. Consequently, our results suggest that U.S. fund companies managing smaller assets tend to outperform those managing larger ones, though this advantage diminishes toward the end of the period. For amc_{it} , the coefficient function is significantly positive in most years, indicating a positive association between higher management charges and fund performance over this period.

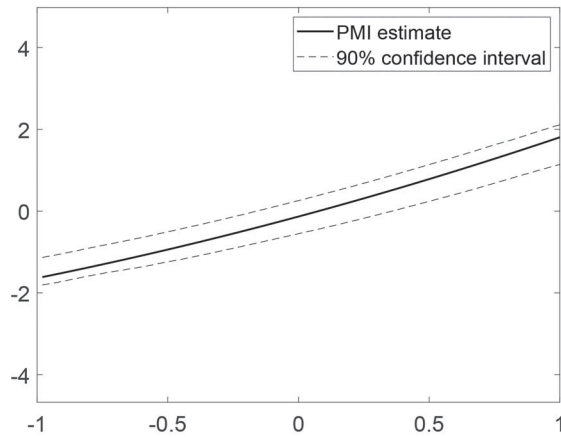
Figure 2 presents the estimated loading functions, which capture fund portfolios' exposure to common risk factors. As can be seen in the figure, the loading functions for value and volatility exhibit evident nonlinearity, suggesting that applying linear factor models to control for common risk factors may yield misleading results due to model misspecification. The confidence intervals indicate that, with the exception of the volatility function, all loading functions are generally significant.



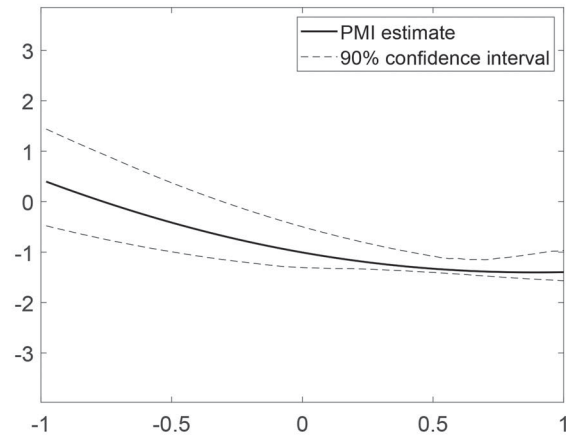
(a) size



(b) value



(c) mom



(d) vol

Figure 2. Estimated loading functions for model (4.1).

4.3. Summary of the Empirical Findings

We now conclude the empirical analysis by addressing the questions (a)–(d) proposed at the end of Section 4.1 using our estimation and testing results. For question (a), our PMI estimates indicate that U.S. mutual fund returns were generally negatively related to the relative size of assets under management before 2017. This suggests that smaller funds produced relatively higher returns during this period. For question (b), mutual fund returns were typically negatively associated with the size of fund family assets, although the estimated coefficient became insignificant by the end of the estimation period. This implies that smaller fund management companies outperformed larger ones in most years, though this superior performance became insignificant after 2018. Combining these findings for questions (a) and (b), we observe that smaller funds in smaller fund families were more likely to earn higher returns in most time periods. For question (c), management charges were generally found to have a significant and positive effect on fund performance, indicating that higher charges are associated with better fund performance.

For question (d), our estimation and test results reveal evident time variation in regression coefficients. This suggests that the relationships between fund size, fund family size, fund management fees, and mutual fund returns may be unstable over time. The test results imply that methods that ignore time variation in coefficients may lead to model misspecification and, consequently, invalid conclusions.

In this section, we have applied the newly proposed methodology to analyze how the mutual fund performance relates to certain fund management information variables while controlling for common risk factors. It is noteworthy that the empirical conclusions presented here focus on associations between variables; neither causality nor predictability was examined in our study.

5. Conclusions and Discussions

This article introduces a nonparametric panel data model that incorporates both time-varying regression coefficients and a partially observed additive factor structure. This flexible model

allows the economic relationships between variables to evolve over time, while capturing the effects of unobservable common factors through nonparametric loading functions. To estimate the unknown quantities, we propose the profile marginal integration (PMI) method, and establish the corresponding asymptotic properties. Additionally, we develop a test to examine the null hypothesis of parametric specifications. An empirical application on U.S. mutual fund returns shows that conventional parametric methods, which overlook time variation, may yield invalid inferences.

While model (1.3) can be used for forecasting after some necessary modifications, the primary interest of this article is to reveal the time-varying patterns in the association between explanatory variables, loading covariates and dependent variables. Therefore, forecasting is left for future research. Another important issue is the factor number selection. For the additive factor structure, developing a variable selection method for observable loading covariates could help determine the number of factors, which we also leave for future exploration.

Supplementary Materials

The supplementary material includes online appendices and replication codes for the empirical study. The online appendices provide detailed proofs of the theorems and offer additional discussions on the proposed methodology.

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The author reports there are no competing interests to declare.

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